TEMPERATURE

Temperature is one of the fundamental quantities in physics and engineering while Heat is a form of energy. Temperature is the measurement of degree of hotness of objects. Two regions in thermal contact have the same temperature when there is no net exchange of internal energy between them. Temperature, then, is what determines the direction of heat flow — *out* of the region with the higher temperature and *in* to the region with the lower temperature. In more concise terms, heat flows from hot to cold.

MEASURMENT OF TEMPERATURE

Temperature is measured with a thermometer. The basic operating principle behind all thermometers is that there is some quantity, called a thermometric variable, that changes in response to changes in temperature. The relationship between temperature and the thermometric variable may be direct or inverse or it may be determined by a polynomial or power function. In any case, it is the thermometric variable that gets measured. There is no way to measure temperature directly.

Types of thermometer

type	thermometric variable
liquid in glass	volume
constant volume gas	pressure
bimetallic strip	coil pitch
electric resistor	resistance
thermocouple	voltage

Once we've settled on the thermometric variable to be measured, the next step is to decide on a temperature scale. In thermometry, what we need are fixed points: reproducible experiments based on natural phenomena that occur at a definite temperature under a proscribed set of conditions. Actually, we need at least two fixed points and a defined range of numbers (called a fundamental interval) between the lower fixed point and the upper fixed point.

Let X stands for any thermometric properties and the temperature θ is directly proportional to X . This linear relation first used in mercury-in-glass thermometer. Thus, the temperature can be given by the following thermometric function

$$\theta(X) = aX \qquad \text{.....(1)}$$

Where a is an arbitrary constant

Equation(1) applies to a thermometer placed in constact with a object whose temperature $\theta(X)$ is to be measured. Therefore, it applies when the thermometer placed in contact with an arbitrarily chosen standard object in reproducible state; such a state of an arbitrary chosen standard object is called a fixed point, that is fixed temperature. The fixed point provides a reference for the determination of temperature scales.

Let X_0 is the value of thermometric property at temperature at $0°C$, X_{100} is at temperature at $100°C$ and also X_θ is at temperature at $\theta°C$ which is to be determined. The mathematical relation at these temperatures with thermometric property

$$\frac{\theta - 0}{100} = \frac{X_\theta - X_0}{X_{100} - X_0} \qquad \text{.....(2)}$$

Problem: Mr. John has made a scale using expansion of metal rod along its length. The rod has length $2m$ at $0°C$ and $2.1m$ at $100°C$. Calculate the temperature corresponding to the length $2.035m$.

TEMPERAURE SCALES

Different types of temperature scales have been established. Among them, most used are:

1. Celsius Scale
2. Fahrenheit Scale
3. Kelvin Scale

Comparative discussion of above scales

Legend	Celsius	Fahrenheit	Kelvin
Sign	$°C$	$°F$	K
Melting point of ice	0	32	273
Boiling point of water	100	212	373

Relation among the temperatures scales:

$$(C - 0)/100 = (F - 32)/(212 - 32) = (K - 273)/(373 - 273)$$

or, $$(C - 0)/100 = (F - 32)/180 = (K - 273)/100$$

or, $$(C - 0)/5 = (F - 32)/9 = (K - 273)/5 \text{ [Dividing by 5]}$$

Problem 1: Find the equivalent value in Celsius scale of 98.4°F .
Problem 2: At what value Celsius and Fahrenheit scales coincide.
Problem 3: Mr. Paul introduced a temperature scale. In his scale, Paul marked $0°C$ as -10 and $100°C$ as 140. Find the equivalent value of 80 in Celsius scale.

Zeroth law of thermodynamics introduces the concept of temperature.

ZEROTH LAW OF THERMODYNAMICS
The zeroth law is a consequence of **thermal equilibrium** and allows us to conclude that temperature is a well-defined physical quantity. The zeroth law of thermodynamics states:
If a body **A** and a body **B** are both in thermal equilibrium with each other; then a body **C** which is in thermal equilibrium with body **B** will also be in equilibrium with body **A** and the temperature of body **C** is equal to the temperature of body **A**(fig. 1).

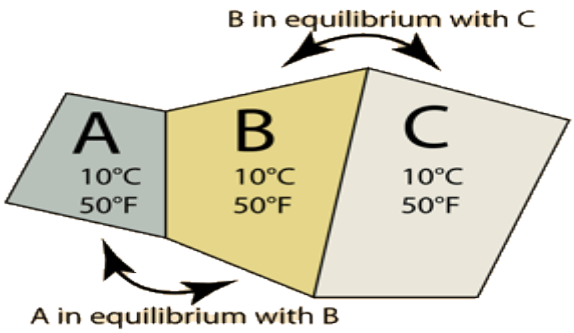


Fig. 1

It is the *zeroth law*, because it preceeds the **first** and **second** laws of thermodynamics and is also a tacit assumption in both laws.
We use the zeroth law when we wish to compare the temperatures of two objects, *A* and *B*. We can do this by using a thermometer, *C* and placing it again object *A* it reaches thermal equilibrium with object *A* and measure the temperature of *A*. Placing the thermometer against object *B* until thermal equilibrium is reached we measure the temperature of object *B*. If they are the same temperature then they will be in thermal equilibrium with each other(fig. 2).

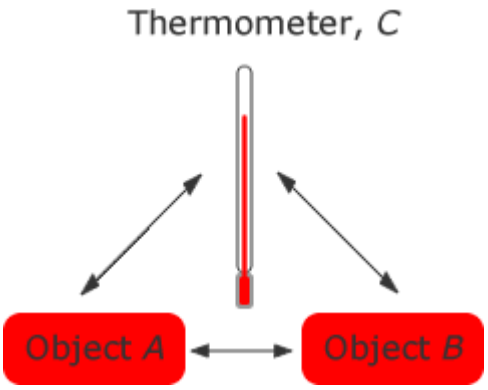


Fig. 2

Thermometers are working examples of the zeroth law of thermodynamics.

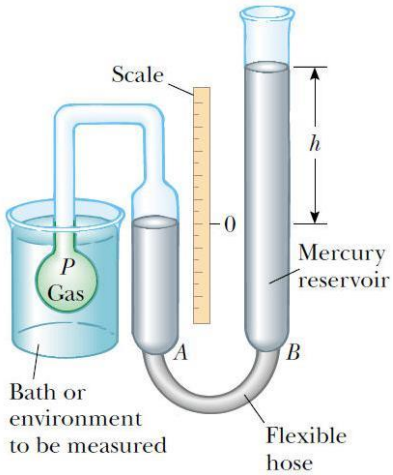
Thermometers

A gas volume thermometer and its usefulness as a standard for other thermometers.

The significance of constant volume gas thermometers is that they are used to calibrate other thermometers.

- Construction

A constant volume gas thermometer is composed of a bulb filled with a fixed amount of a dilute gas that is attached to a mercury manometer. A manometer is a device used to measure pressure.



The mercury manometer has a column partially filled with mercury that is connected to a flexible tube that has another partially filled column of mercury, called a reservoir, attached to the other end. The height of the mercury in the first column is set to a reference point or pressure P that it must stay at, while the mercury in the reservoir is allowed to move up and down in relation to a scale or ruler.

From the Law of Gay-Lussac, we know that when the temperature of an ideal gas increases, that there is a corresponding increase in pressure. Conversely, when the temperature decreases, so does the pressure. Constant volume gas thermometers operate on the same principle, with the restrictions that the gas is at low pressure and the temperature of it is well above that of liquefaction.

Calculations

For a pressure P, the equation used to find the temperature T is

1) $T = aP + b$

where a and b are constants determined from two fixed points, such as ice at 0°C and steam at 100°C.

As mentioned previously, the constant volume gas thermometer uses values from the triple point of water to calibrate other thermometers. To recall, 273.16 K (Kelvin) is the temperature where water exists in an equilibrium state as a gas, liquid, and solid. For equation 1, where a = 273.16 K, b = 0, P_{tp} is the pressure of the gas at the triple point of water, and P is the pressure of the gas at the temperature to be measured, we have

2) $T = 273.16\ K\ (P/P_{tp})$

For low pressure and high temperatures, where real gases behave like ideal gases, equation 1 becomes:

3) $T = 273.16\ K \lim P/P_{tp}$ as $P_{tp} \rightarrow 0$

Operation
The gas bulb is inserted into a bath or place we wish to find the temperature of, such as water.

The pressure of the mercury also changes such that it begins to move up or down and thus away from the reference point. To stop this movement, which will also stop the gas from expanding, the reservoir at the other end is physically lifted up or down, and the ensuing new pressure at this height is measured. The difference between the reference and reservoir heights gives the final pressure P, which is then used to calculate the temperature.
Advantages: very accurate, has a wide range, is very sensitive and is independent of the gas used. Disadvantages: large, cumbersome and slow.
By using different gas thermometers a wide range of temperatures can be measured:

Hydrogen -200 °C to +500 °C
Nitrogen +500 °C to + 1500 °C
Helium -270 °C to + 1500 °C

These thermometers can be very accurate, to within 0.005 °C from 0 °C to 100 °C, 0.1 °C around 500 °C and to within 2 °C at 1500 °C

RESISTANCE THERMOMETERS
Resistance thermometers, also called **resistance temperature detectors** (**RTDs**), are sensors used to measure temperature. Many RTD elements consist of a length of fine wire wrapped around a ceramic or glass core but other constructions are also used. RTDs have higher accuracy and repeatability, are slowly replacing thermocouples in industrial applications below 600 °C.

Common RTD sensing elements constructed of platinum, copper or nickel have a repeatable resistance versus temperature relationship (*R* vs *T*) and operating temperature range. The *R* vs *T* relationship is defined as the amount of resistance change of the sensor per degree of temperature change. The relative change in resistance (temperature coefficient of resistance) varies only slightly over the useful range of the sensor.

Platinum was proposed by Sir William Siemens as an element for resistance temperature detector at the Bakerian lecture in 1871: it is a noble metal and has the most stable resistance–temperature relationship over the largest temperature range. Nickel elements have a limited temperature range because the amount of change in resistance per degree of change in temperature becomes very non-linear at temperatures over 572 °F (300 °C). Copper has a very linear resistance–temperature relationship; however, copper oxidizes at moderate temperatures and cannot be used over 302 °F (150 °C).
Platinum is the best metal for RTDs due to its very linear resistance–temperature relationship, highly repeatable over a wide temperature range. The unique properties of platinum make it the material of choice for temperature standards over the range of −272.5 °C to 961.78 °C. Platinum is chosen also because of its chemical inertness.
The significant characteristic of metals used as resistive elements is the linear approximation of the resistance versus temperature relationship between 0 and 100 °C. This temperature coefficient of resistance is denoted by α and is usually given in units of $\Omega/(\Omega \cdot {}^\circ C)$:

$$a = \frac{R_{100} - R_0}{100 R_0}$$

Where

R_0 is the resistance of the sensor at 0 °C,

R_{100} is the resistance of the sensor at 100 °C.
Pure platinum has a = 0.003925 Ω/(Ω·°C) in the 0 to 100 °C range and is used in the construction of laboratory-grade RTDs. Conversely, two widely recognized

CONSTRUCTION:

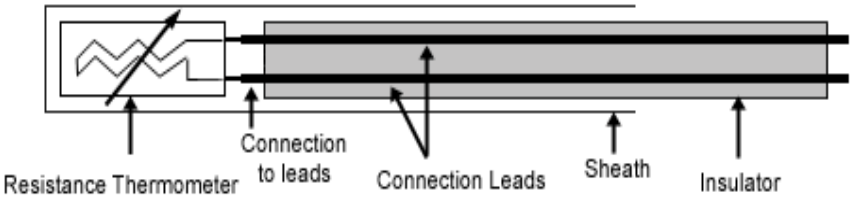


Fig. 2

These elements nearly always require insulated leads attached. PVC, silicone or rubber insulators are used at temperatures below about 250 °C. Above this, glass fibre or ceramic are used. The measuring points, and usually most of the leads, require a housing or protective sleeve, often made of a metal alloy that is chemically inert to the process being monitored. Selecting and designing protection sheaths can require more care than the actual sensor, as the sheath must withstand chemical or physical attack and provide convenient attachment points.
The calibration of the instrument often involves the following quadratic equation:

$$\boldsymbol{R_T = R_r(1 + aT + bT^2)} \qquad (4)$$

Where $\boldsymbol{R(T)}$ is the resistance of the platinum wire at the temperature $\boldsymbol{T}$, $\boldsymbol{R_r}$ is the resistance of the platinum wire at the known point, and $\boldsymbol{a}$ and $\boldsymbol{b}$ are constants.

THERMOCOUPLE:

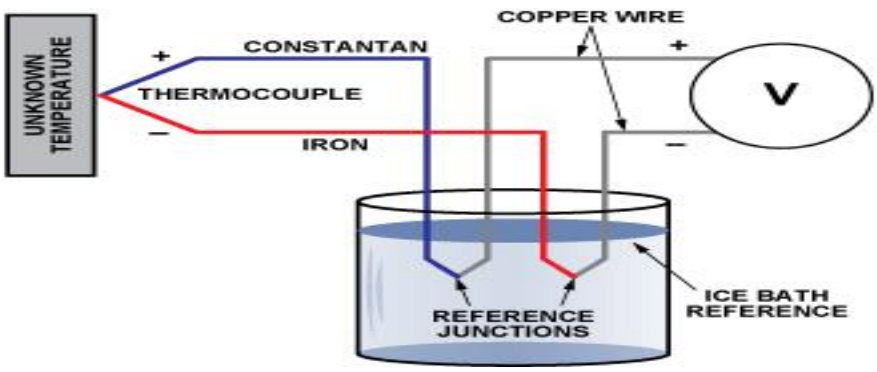
A thermocouple is an electrical device consisting of two dissimilar conductors forming electrical junctions at differing temperatures. A thermocouple produces a temperature-dependent voltage as a result of the thermoelectric effect, and this voltage can be interpreted to measure temperature. Thermocouples are a widely used type of temperature sensor.

Commercial thermocouples are inexpensive, interchangeable, are supplied with standard connectors, and can measure a wide range of temperatures. In contrast to most other methods of temperature measurement, thermocouples are self-powered and require no external form of excitation. The main limitation with thermocouples is accuracy; system errors of less than one degree Celsius (°C) can be difficult to achieve.

Thermocouples are widely used in science and industry; applications include temperature measurement for kilns(a type of oven, that produces temperatures sufficient to complete some process, such as hardening, drying, or chemical changes. Kilns have been used for millennia to turn objects made from clay into pottery, tiles and bricks), gas turbine exhaust, diesel engines, and other industrial processes. Thermocouples are also used in homes, offices and businesses as the temperature sensors in thermostats, and also as flame sensors in safety devices for gas-powered major appliances.

PRINCIPLE OF OPERATION

In 1821, the German physicist Thomas Johann Seebeck discovered that when different metals are joined at the ends and there is a temperature difference between the joints, a magnetic field is observed. At the time Seebeck referred to this as thermo-magnetism. The magnetic field he observed was later shown to be due to thermo-electric current. In practical use, the voltage generated at a single junction of two different types of wire is what is of interest as this can be used to measure temperature at very high and low temperatures. The magnitude of the voltage depends on the types of wire used. Generally, the voltage is in the microvolt range and care must be taken to obtain a usable measurement. Although current flows very little, power can be generated by a single thermocouple junction. Power generation using multiple thermocouples, as in a thermopile, is common.



A schematic configuration for thermocouple usage is shown in the figure. Briefly, the desired temperature T_{sense} is obtained using three inputs—the characteristic function $E(T)$ of the thermocouple, the measured voltage V, and the reference junctions' temperature T_{ref}. The solution to the equation $E(T_{sense}) = V + E(T_{ref})$ yields T_{sense}.
Type K (chromel–alumel) is the most common general-purpose thermocouple with a sensitivity of approximately 41 µV/°C. It is inexpensive, and a wide variety of probes are available in its −200 °C to +1350 °C (−330 °F to +2460 °F) range.

PYROMETER

A pyrometer is a type of remote-sensing thermometer used to measure the temperature of a surface. Various forms of pyrometers have historically existed. In the modern usage, it is a device that from a distance determines the temperature of a surface from the spectrum of the thermal radiation it emits, a process known as pyrometry and sometimes radiometry.

The word pyrometer comes from the Greek word for fire, "πυρ" (*pyro*), and *meter*, meaning to measure. The word pyrometer was originally coined to denote a device capable of measuring the temperature of an object by its incandescence, visible light emitted by a body which is at least red-hot. Modern pyrometers or infrared thermometers also measure the temperature of cooler objects, down to room temperature, by detecting their infrared radiation flux.

A modern pyrometer has an optical system and a detector. The optical system focuses the thermal radiation onto the detector. The output signal of the detector (temperature T) is related to the thermal radiation or irradiance j^* of the target object through the Stefan–Boltzmann law, the constant of proportionality σ, called the Stefan-Boltzmann constant and the emissivity ε of the object.

$$j^* = \varepsilon \sigma T^4$$

This output is used to infer the object's temperature from a distance, with no need for the pyrometer to be in thermal contact with the object; most other thermometers (e.g. thermocouples and resistance temperature detectors (RTDs)) are placed in thermal contact with the object, and allowed to reach thermal equilibrium. Pyrometry of gases presents difficulties. These are most commonly overcome by using thin filament pyrometry or soot pyrometry. Both techniques involve small solids in contact with hot gases.